

# Lewis structure of methane, CH<sub>4</sub>

## Strategy:

- 1) Write skeletal structure of molecule
- 2) Count the total number of valence electrons
- 3) Draw all bonds as single bonds and fill remaining electrons to complete octets
- 4) If it does not add up, use lone-pairs to generate double or triple bonds.

## Exercise:

Write Lewis structures for:

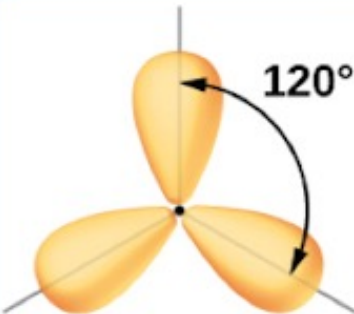
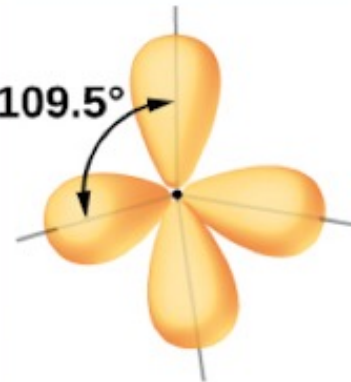
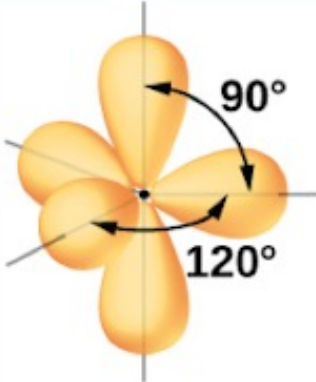
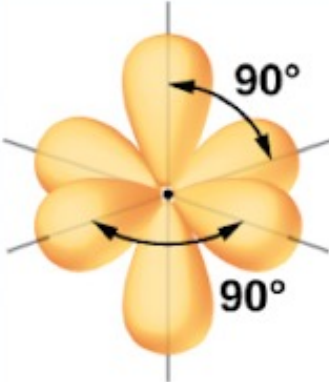
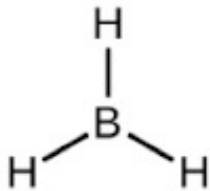
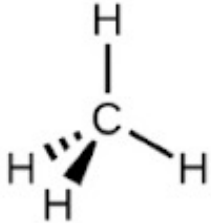
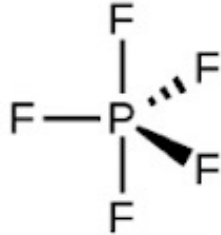
- a) N<sub>2</sub>
- b) NH<sub>3</sub>
- c) CH<sub>3</sub>CH<sub>2</sub>OH
- d) BH<sub>3</sub>

| Group<br>Period | 1        | 2        | 3        | 4         | 5         | 6         | 7         | 8         | 9         | 10        | 11        | 12        | 13        | 14        | 15        | 16        | 17        | 18        |           |
|-----------------|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| 1               | 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |           |
| 2               | 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           | 5<br>B    | 6<br>C    | 7<br>N    | 8<br>O    | 9<br>F    | 10<br>Ne  |           |
| 3               | 11<br>Na | 12<br>Mg |          |           |           |           |           |           |           |           |           |           | 13<br>Al  | 14<br>Si  | 15<br>P   | 16<br>S   | 17<br>Cl  | 18<br>Ar  |           |
| 4               | 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |           |
| 5               | 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |           |
| 6               | 55<br>Cs | 56<br>Ba | *        | 71<br>Lu  | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |
| 7               | 87<br>Fr | 88<br>Ra | *        | 103<br>Lr | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Mc | 116<br>Lv | 117<br>Ts | 118<br>Og |
|                 |          |          | *        | 57<br>La  | 58<br>Ce  | 59<br>Pr  | 60<br>Nd  | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  |           |           |
|                 |          |          | *        | 89<br>Ac  | 90<br>Th  | 91<br>Pa  | 92<br>U   | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No |           |           |

# VSEPR Model

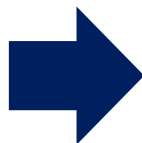
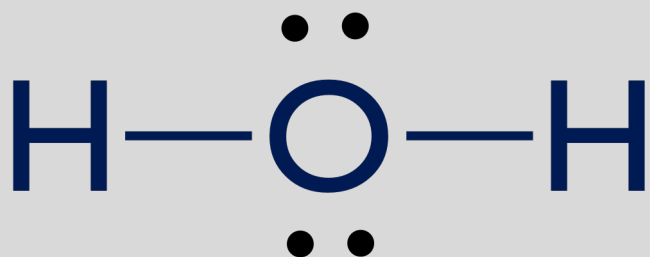
- **V**alence **S**hell **E**lectron **P**air **R**epulsion model – simple method to predict the geometry of (most) molecules.
- The idea: count the number of electron pairs around an atom. The electron pair will maximize the distance between them - i.e. each electron pair have as much room around them as possible.
- Whether the electron pair is a chemical bond or a lone pair does not matter



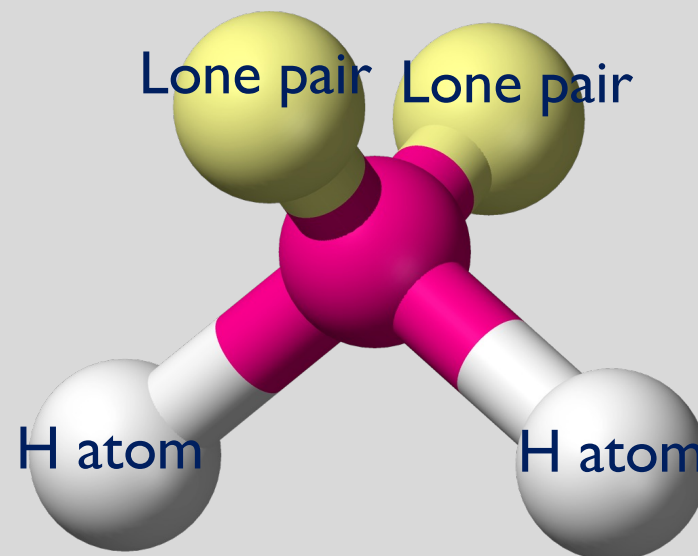
|                                                                                   |                                                                                    |                                                                                      |                                                                                            |                                                                                      |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |   |    |         |   |
| $\text{H}-\text{Be}-\text{H}$                                                     |  |  |        |  |
| <p>Linear;<br/><math>180^\circ</math> angle</p>                                   | <p>Trigonal planar;<br/>all angles <math>120^\circ</math></p>                      | <p>Tetrahedral;<br/>all angles <math>109.5^\circ</math></p>                          | <p>Trigonal bipyramidal;<br/>angles of <math>90^\circ</math> or <math>120^\circ</math></p> | <p>Octahedral;<br/>all angles <math>90^\circ</math> or <math>180^\circ</math></p>    |

# VSEPR Example I: $\text{H}_2\text{O}$

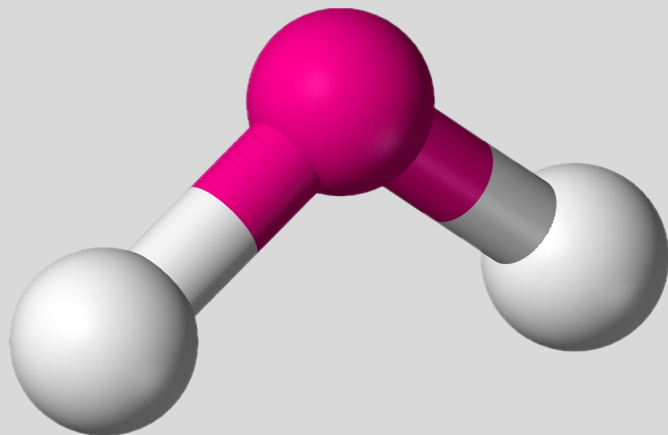
Write Lewis structure:



4 electron densities to arrange in space: tetrahedral arrangement



Final structure: **bent**

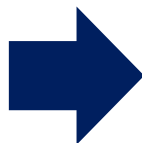
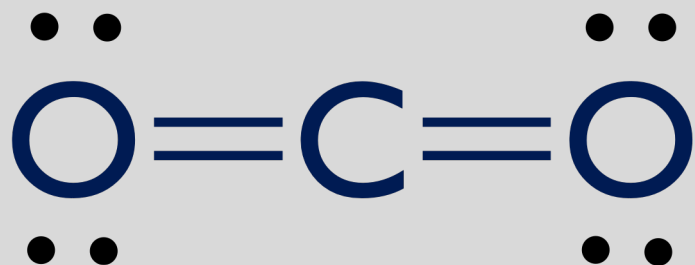


H—O—H angle *ca.* 105 deg.

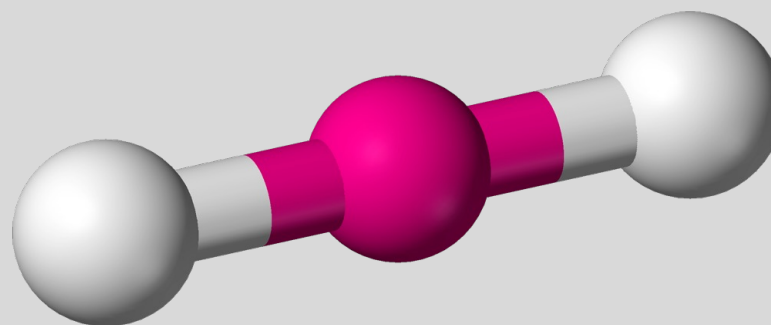
Show that water is a polar molecule!

# VSEPR Example II: CO<sub>2</sub>

Write Lewis structure:



2 electron densities to arrange in space: tetrahedral arrangement

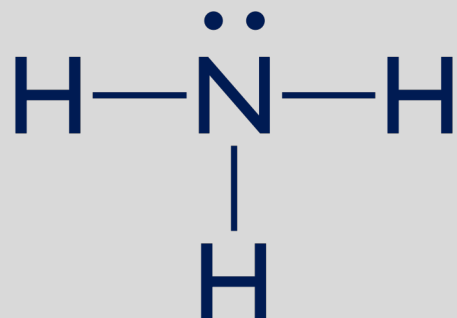


Final structure: **linear**

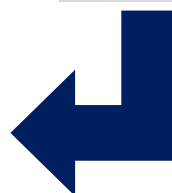
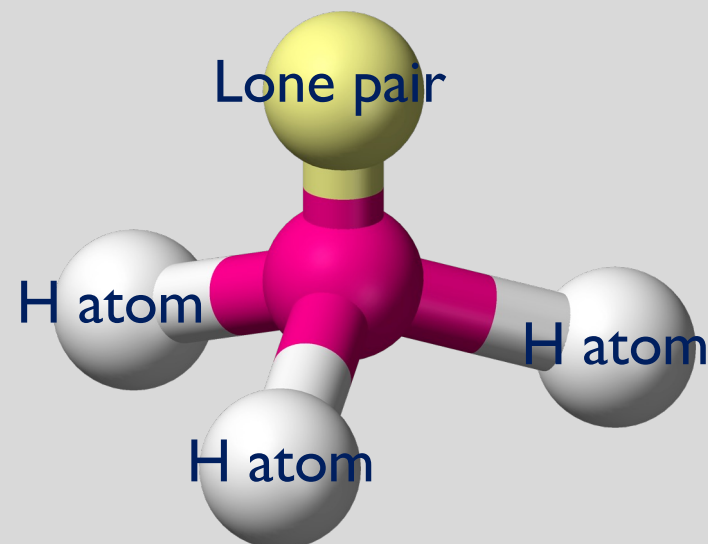
Is CO<sub>2</sub> a polar molecule?

# VSEPR Example III: $\text{NH}_3$

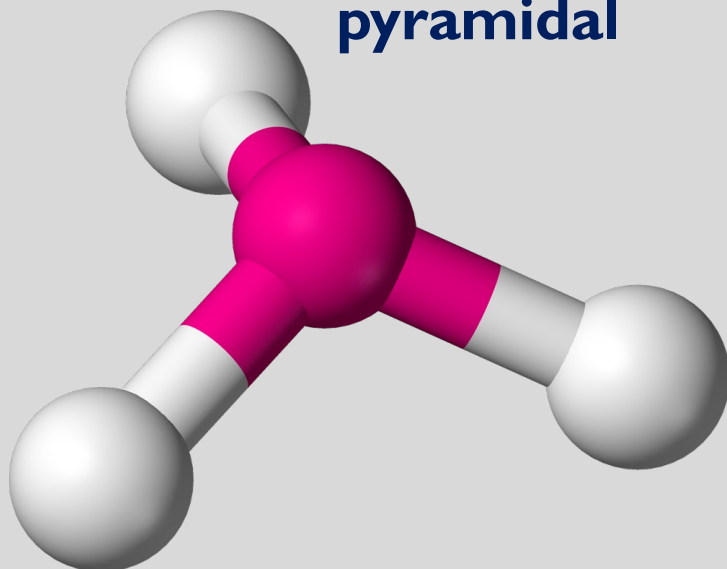
Write Lewis structure:



4 electron densities to arrange in space: tetrahedral arrangement

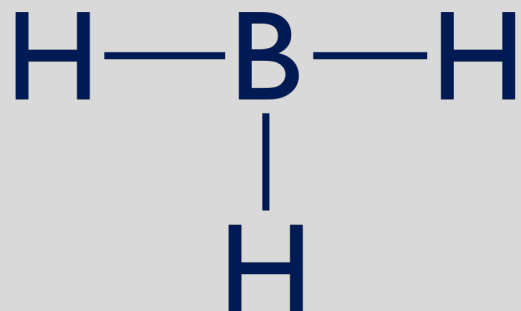


Final structure: **trigonal pyramidal**

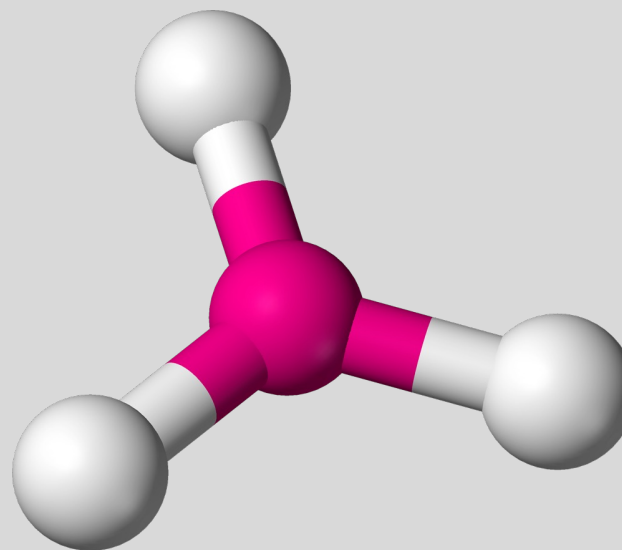


# VSEPR Example IV: $\text{BH}_3$

Write Lewis structure:



3 electron densities to arrange in space: trigonal arrangement

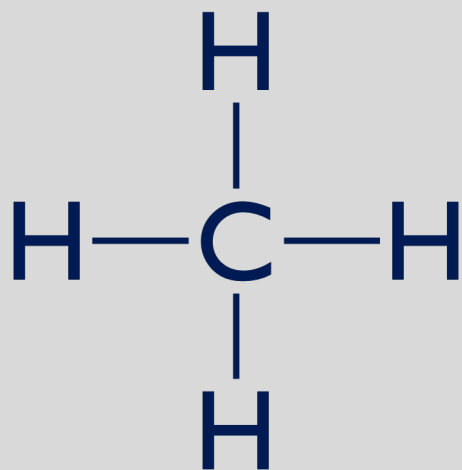


Final structure: **trigonal planar**

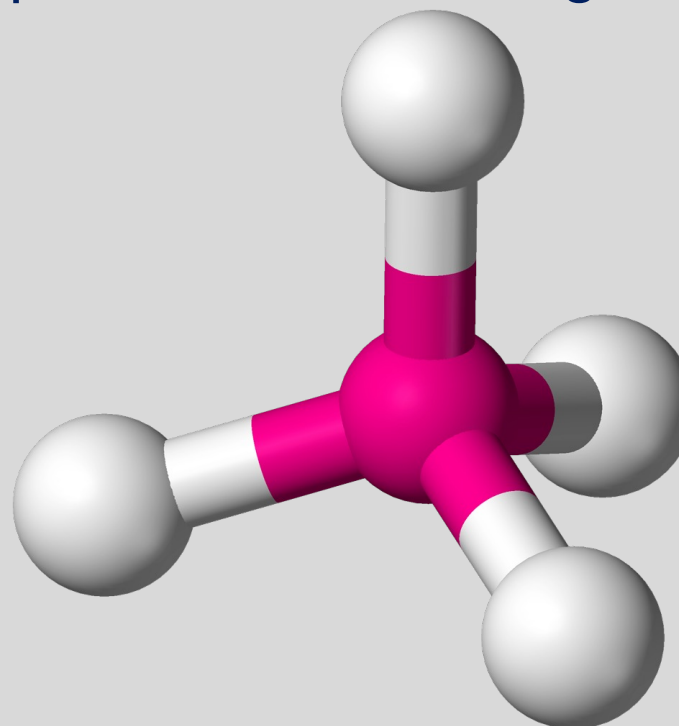


# VSEPR Example V: CH<sub>4</sub>

Write Lewis structure:



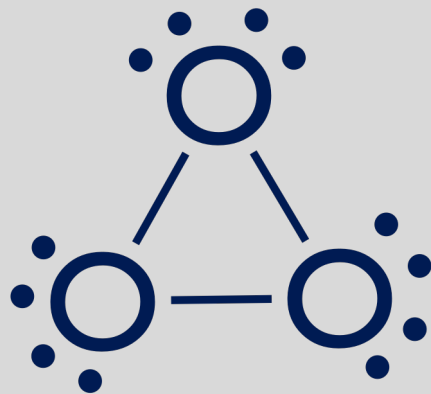
4 electron densities to arrange in space: tetrahedral arrangement



Final structure: **tetrahedral**

# Inadequacy of Lewis structures: Ozone, $O_3$

Write Lewis structure:

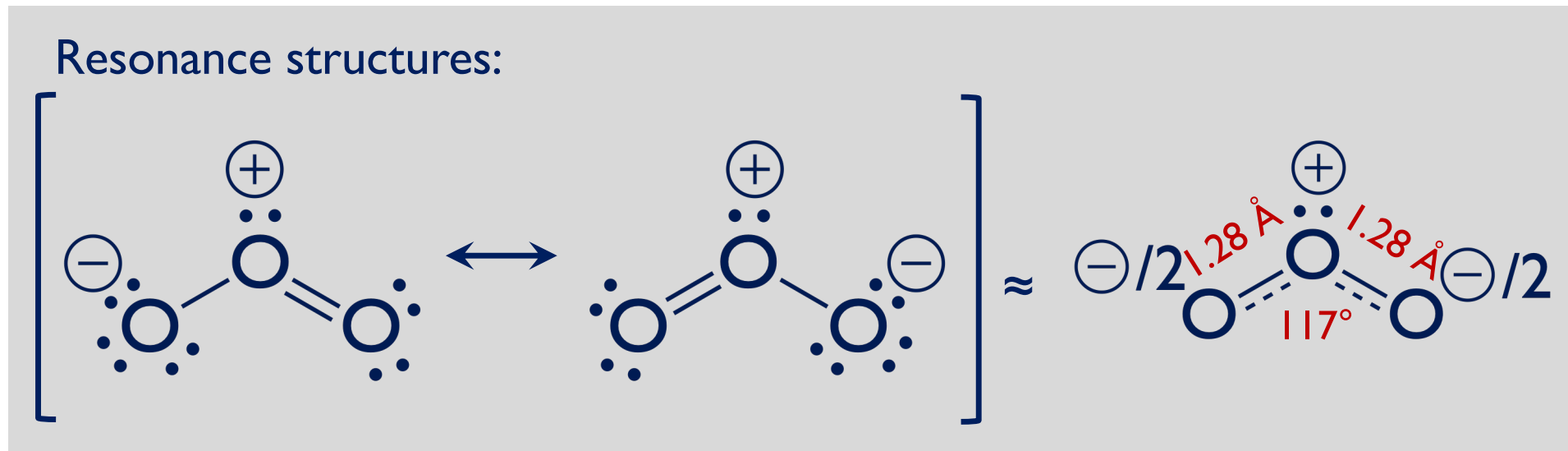


**Problem:** the experimentally determine  $O-O-O$  bond angle is  **$117^\circ$** , very far from the expected  **$60^\circ$** !

Still, all  $O-O$  bond lengths are identical ( $1.28 \text{ \AA}$ )

# Inadequacy of Lewis structures:

## Ozone

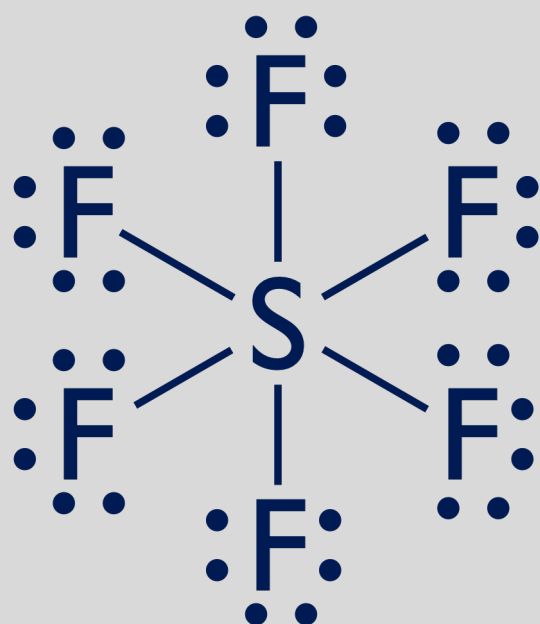


A resonance *hybrid*: the electronic ground state best described as several Lewis structures termed *resonance forms*. Real electronic structure - a mixture of the resonance forms.

**Often** the resonance form that carries the fewest charges and where any negative charges are localized on the more electronegative elements is favored.

# VSEPR Example VI: $\text{SF}_6$ and the expanded octet

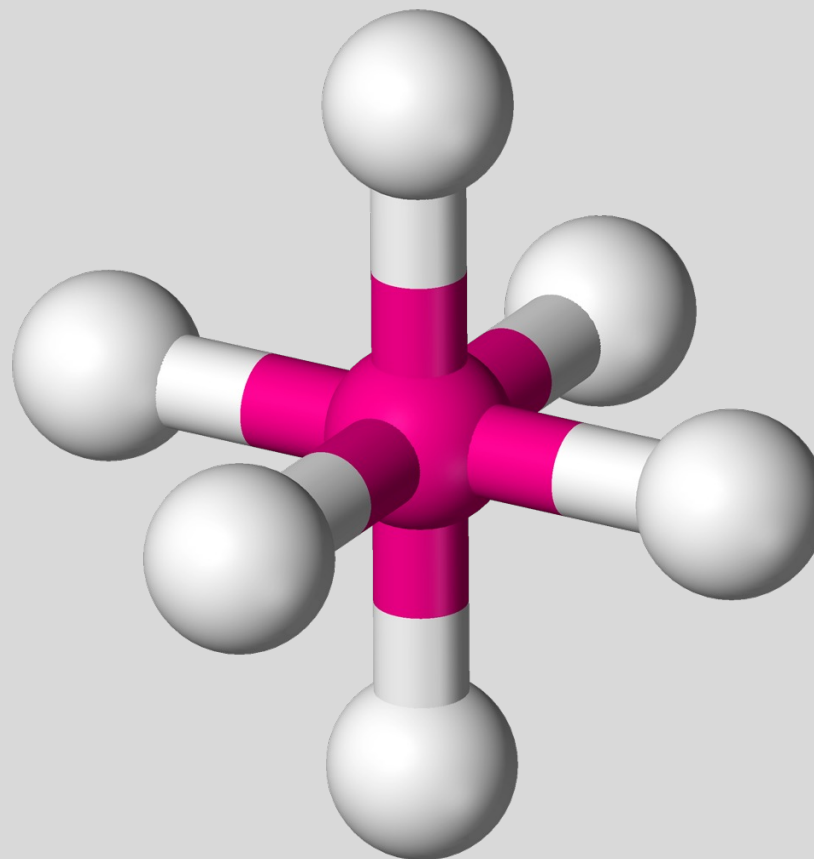
Write Lewis structure:



Sulfur surrounded by 12  $e^-$ !?  
OK – *not 2<sup>nd</sup> period*



6 electron densities to arrange in space: octahedral arrangement



# More exercises

Write Lewis structures (incl. formal charges) and predict structures using the VSEPR model for:

- a)  $\text{IF}_5$
- b)  $\text{SOCl}_2$
- c)  $\text{ClO}_3^-$
- d)  $\text{SF}_4$
- e)  $\text{CH}_3\text{OH}$
- f)  $\text{XeF}_2$
- g)  $\text{NO}_2^-$

## Strategy:

- 1) Write skeletal structure of molecule
- 2) Count the total number of valence electrons
- 3) Draw all bonds as single bonds and fill remaining electrons to complete octets
- 4) If it does not add up, use lone-pairs to generate double or triple bonds.

# Intermolecular forces and liquids and solids

26021 Chemistry (Polytechnical Foundation)

Professor Kasper S. Pedersen

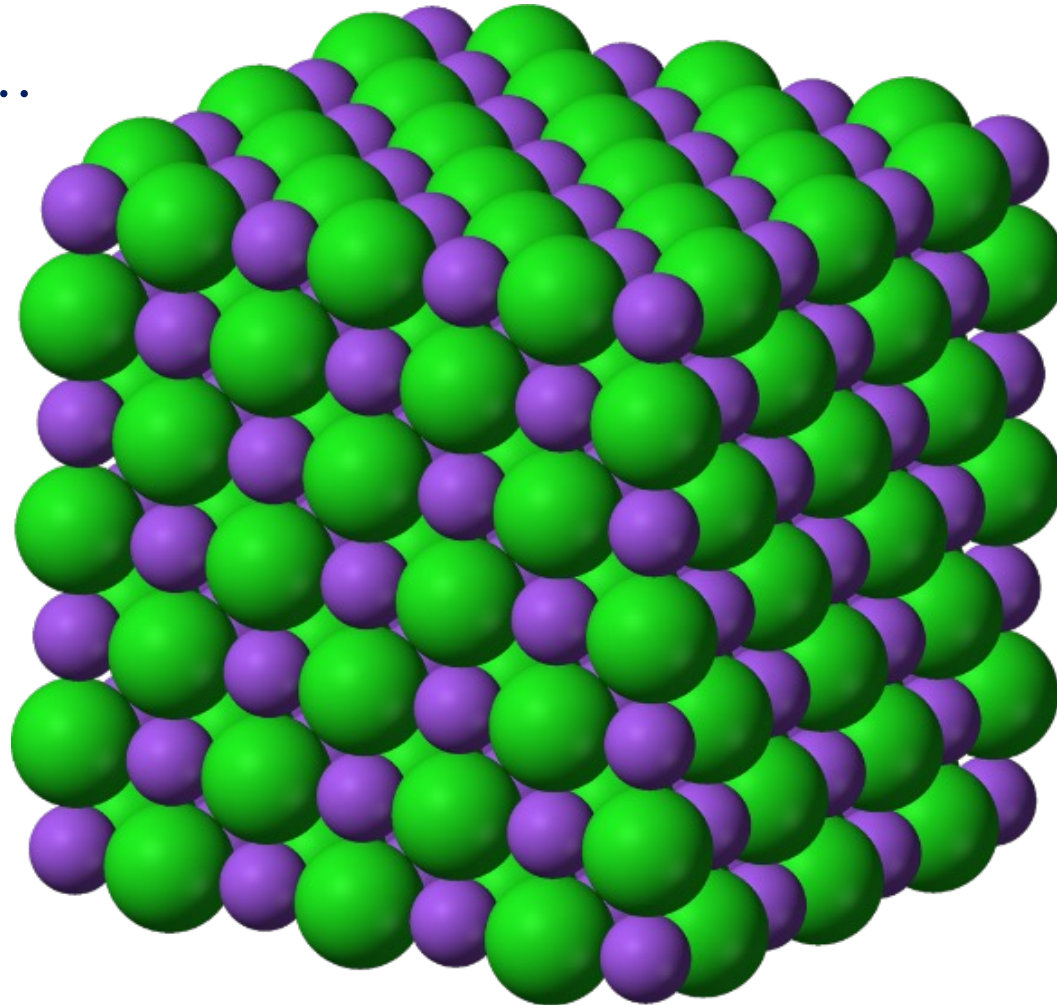
2026

# Ionic bonds

Electrons are localized – thus low electric conductivity.

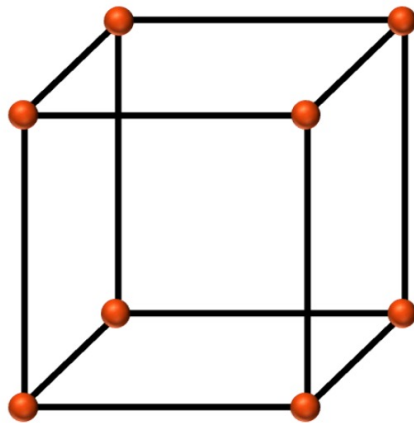
Bond energy  **$\sim 400\text{-}4000$  kJ/mol**; electrostatic interaction (Coulomb's Law)

E.g. MgO, NaCl,...

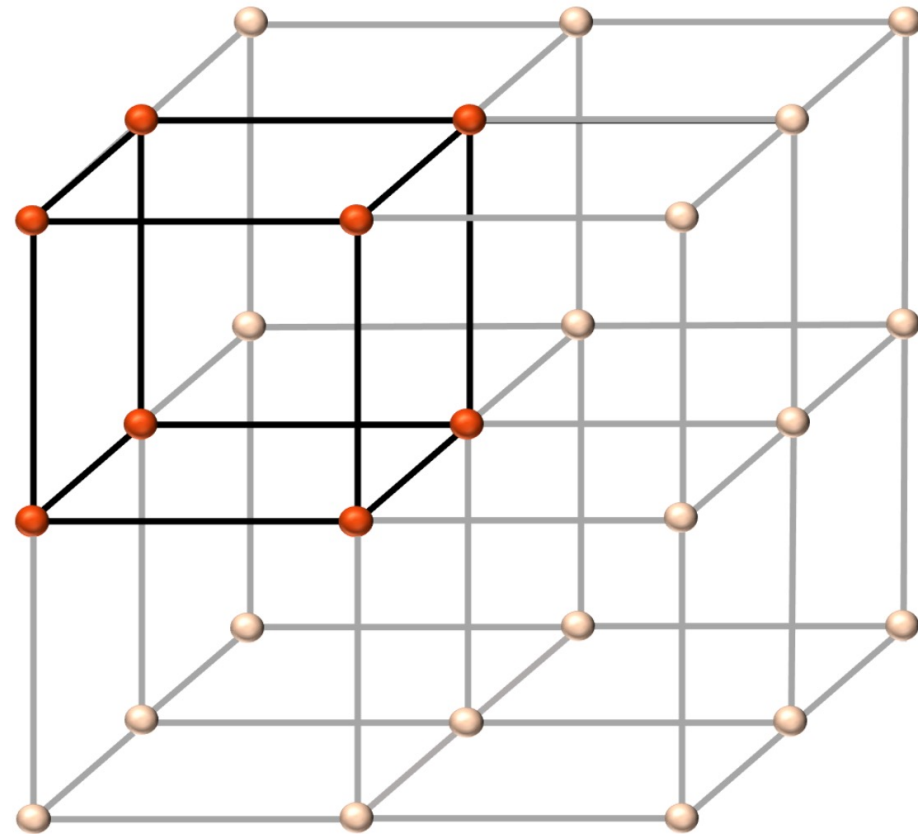


# Crystals

In the description of a crystal replace the repeating unit – e.g. water in ice – with a point. Organize the point in cells such that all point can be described by translating a small unit – the unit cell:



Unit cell

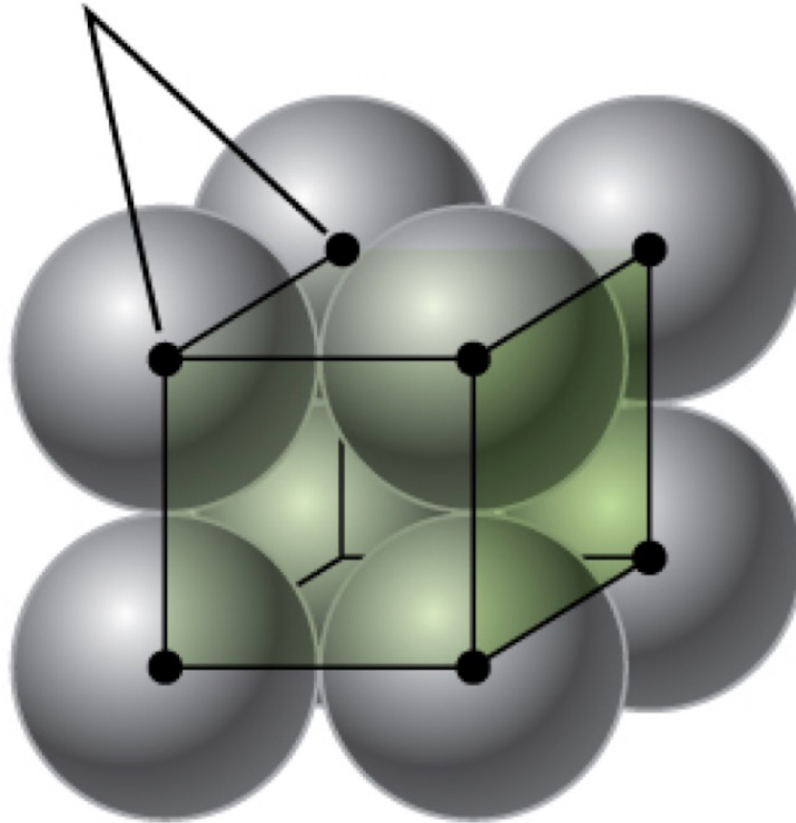


Crystal lattice



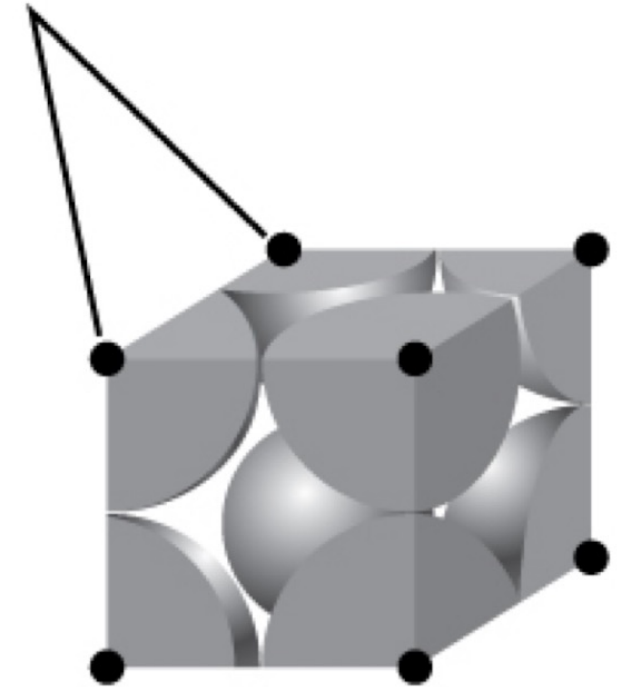
# Crystals

Lattice points



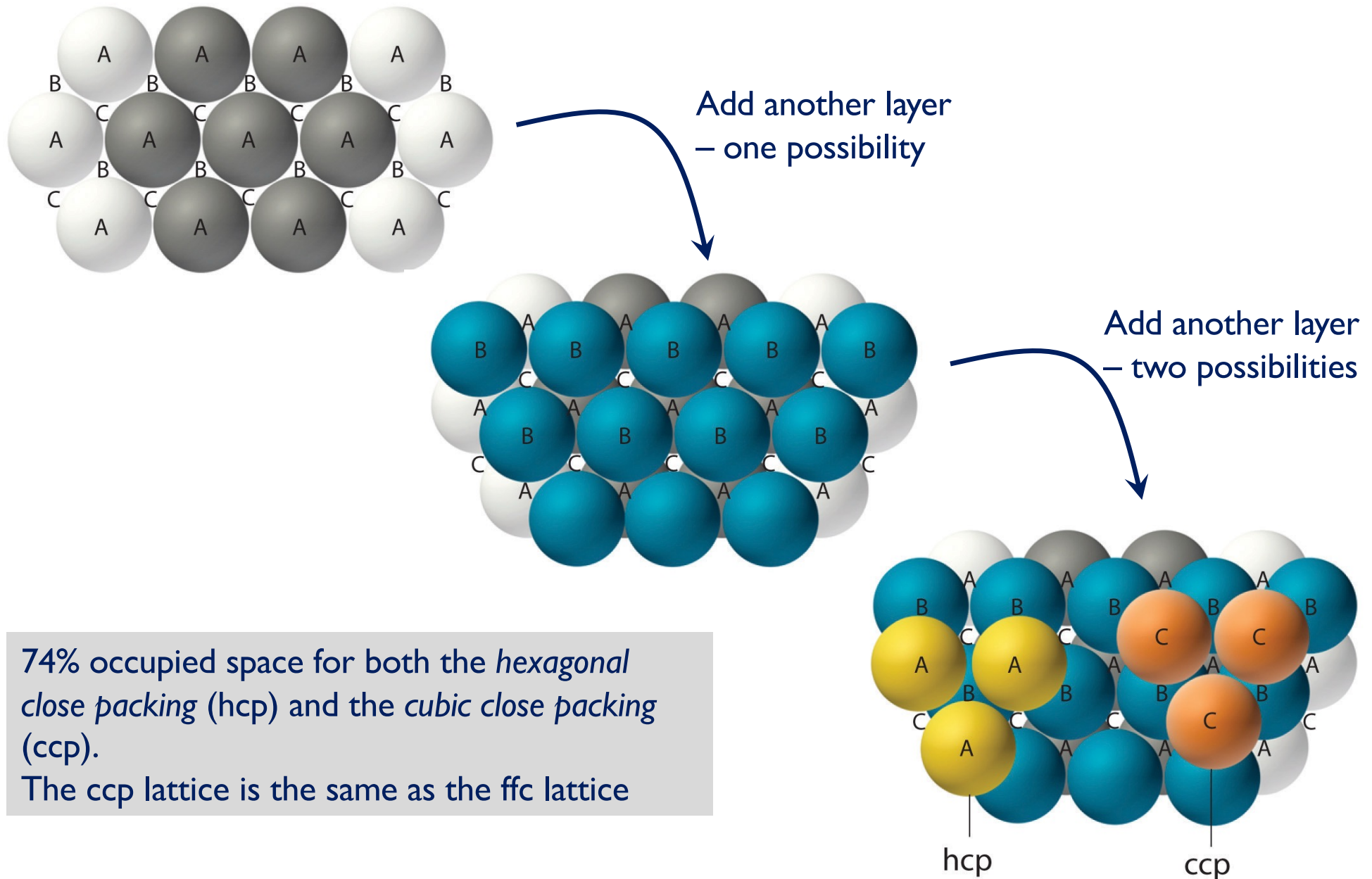
Simple cubic lattice cell

Lattice points



8 corners

# Close packing of spheres

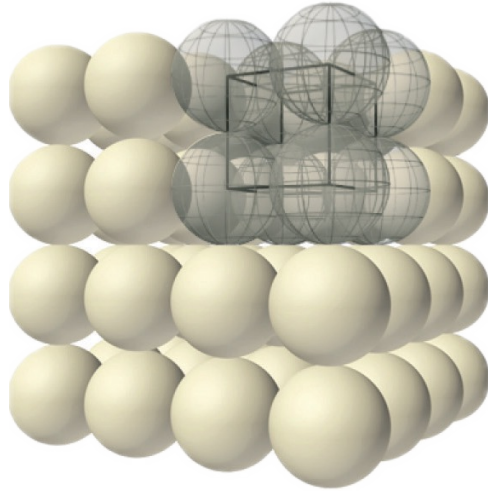


74% occupied space for both the *hexagonal close packing* (hcp) and the *cubic close packing* (ccp).

The ccp lattice is the same as the fcc lattice

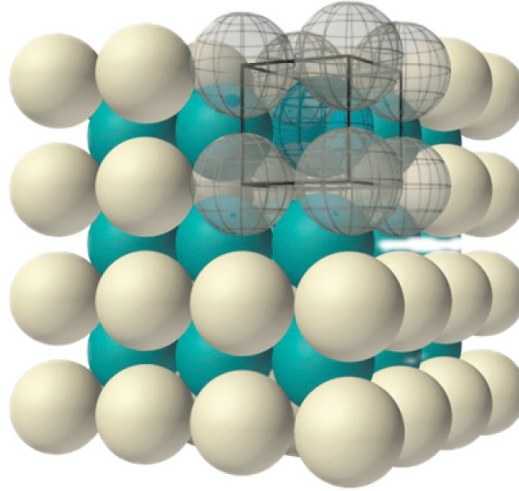
# Packing spheres

Simple cubic



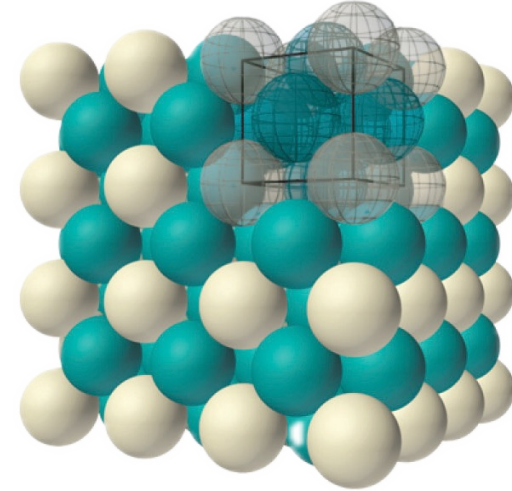
52% occupied space

Body-centered cubic



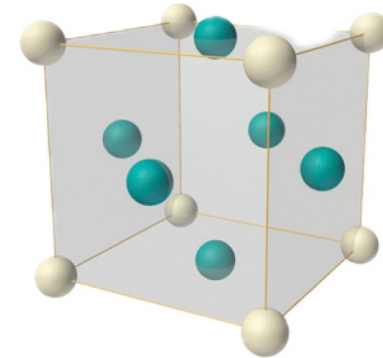
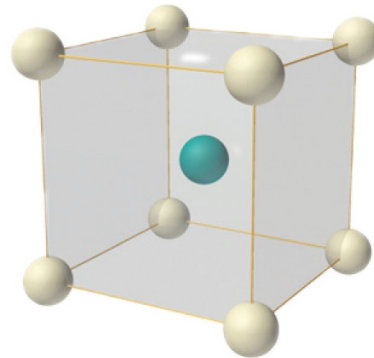
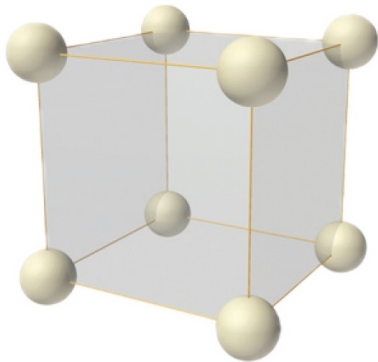
68% occupied space

Face-centered cubic



74% occupied space

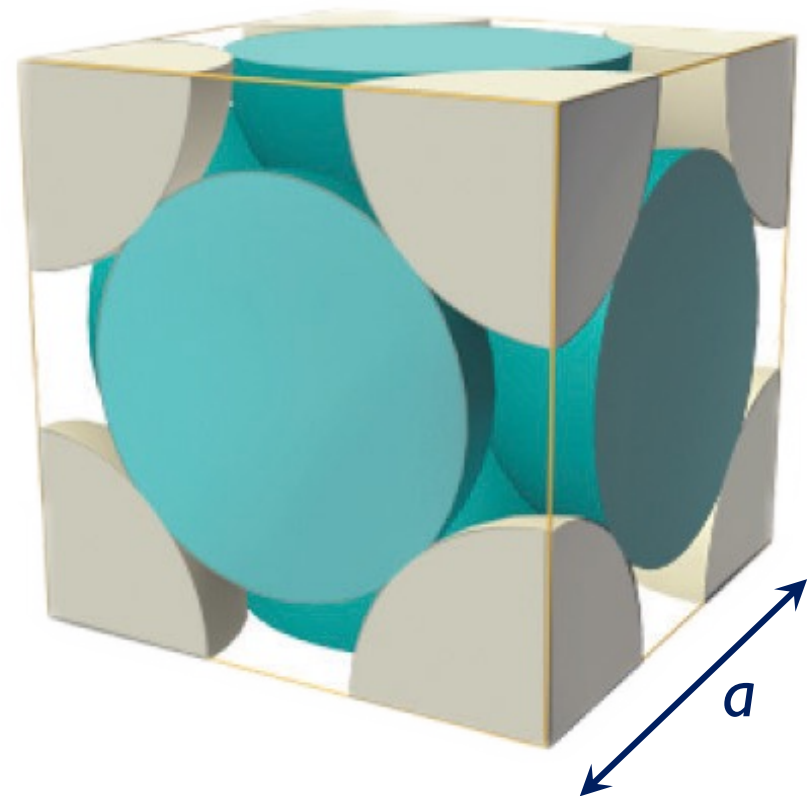
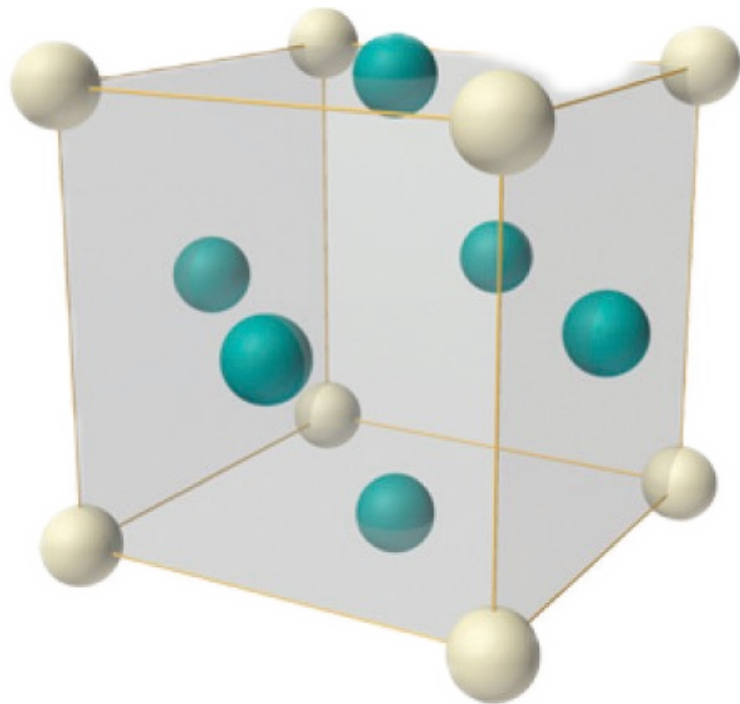
Depiction of unit cells:



# Exercise

Gold crystallizes in the face-centered cubic cell  
The density of gold is  $19.3 \text{ g cm}^{-3}$ .

How many atoms are in the unit cell?  
What is the atomic radius of Au?



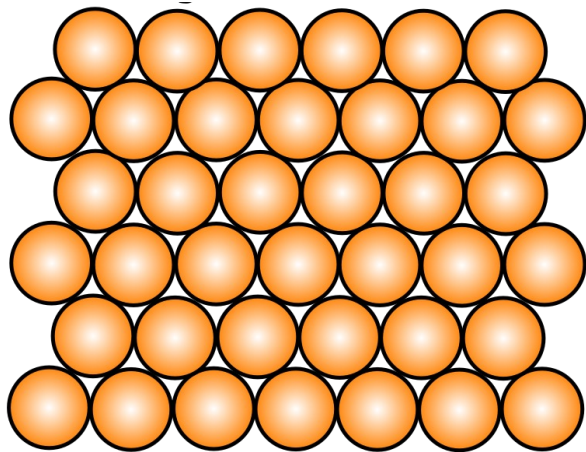


# Amorphous materials

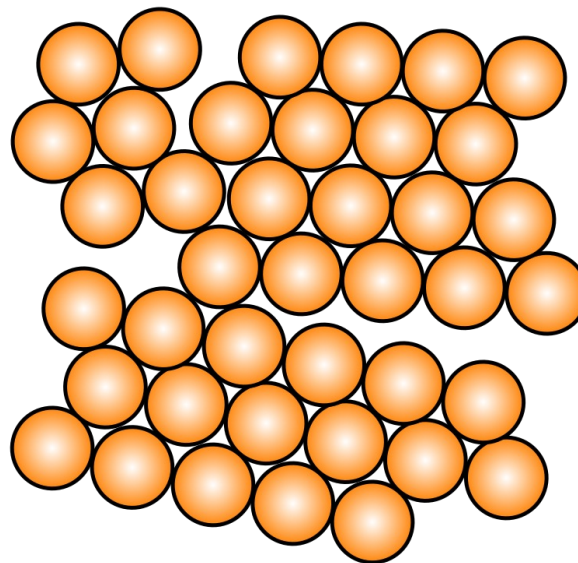
No long range order, i.e. no periodicity in the position of the atoms

**Liquid:** conforms to the container

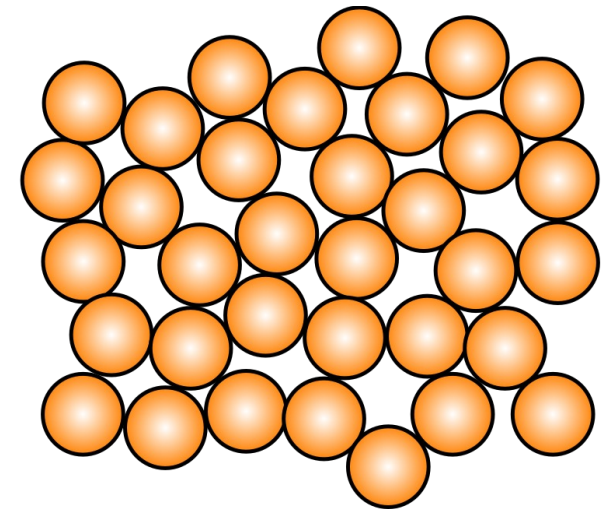
**Solid:** do not flow perceptibly under moderate stress  $\Rightarrow$  glasses



Crystalline



Polycrystalline



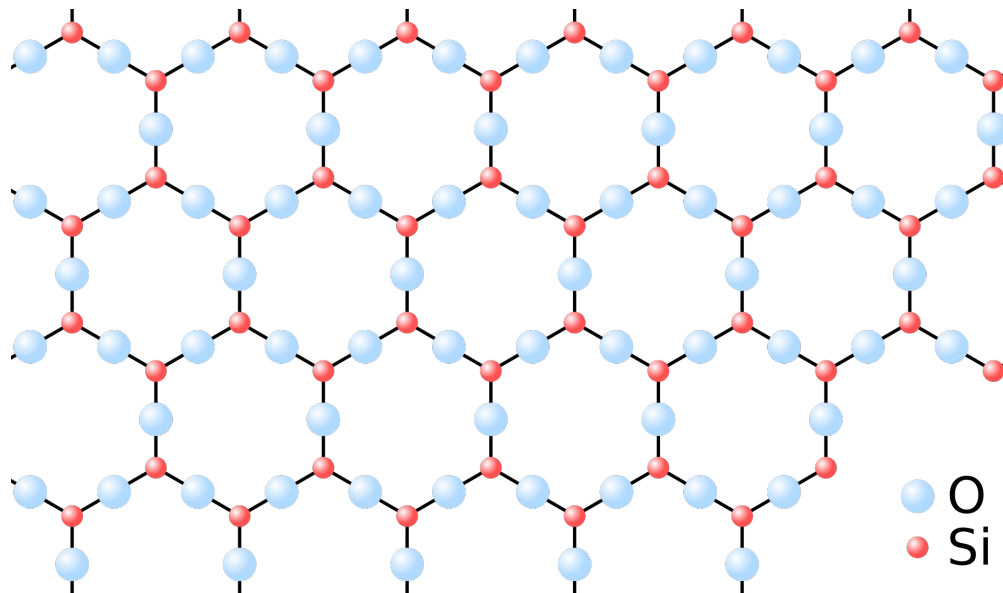
Amorphous

# Amorphous materials

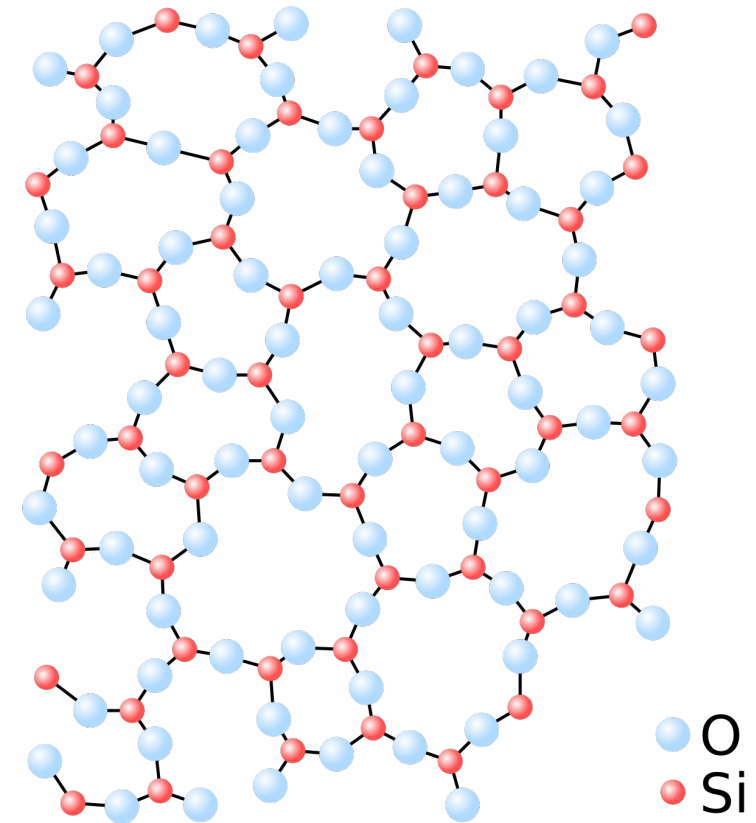
No long range order, i.e. no periodicity in the position of the atoms

**Liquid:** conforms to the container

**Solid:** do not flow perceptibly under moderate stress  $\Rightarrow$  glasses



$\text{SiO}_2$  (tridymite)



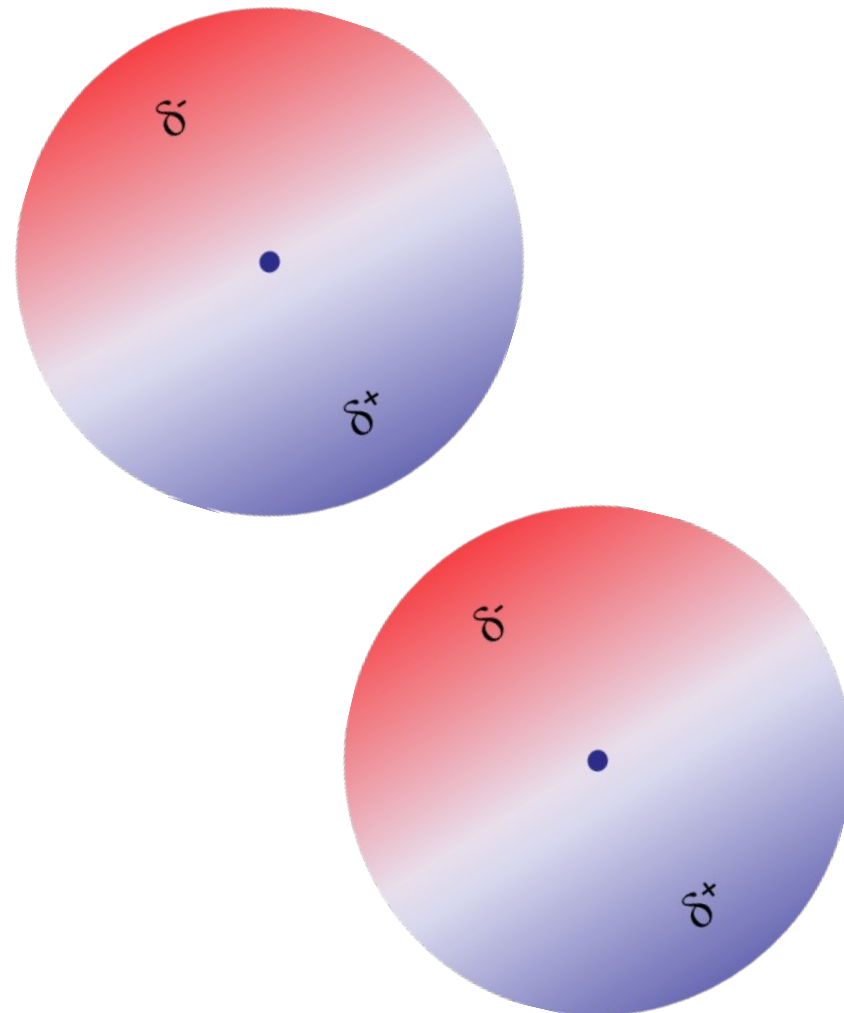
$\text{SiO}_2$  (glass)

# Van der Waals interactions

*Weakest* chemical bonds: between permanent dipole, permanent dipoles and induced dipole, induced dipoles.

**Bond energy < 10 kJ/mol**

E.g.: noble gas liquids and crystals (He, Ne, Ar, ...), interactions between molecules



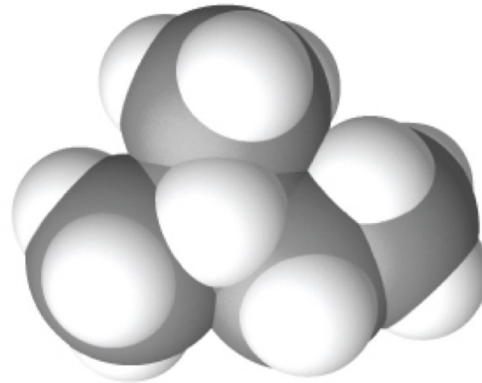
# Van der Waals interactions

Small contact area,  
weakest attraction



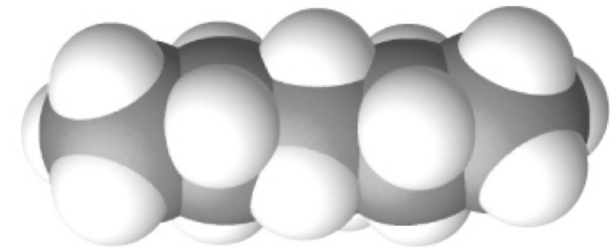
neopentane  
boiling point: 9.5 °C

Less surface area,  
less attraction



isopentane  
boiling point: 27 °C

Large contact area,  
strong attraction



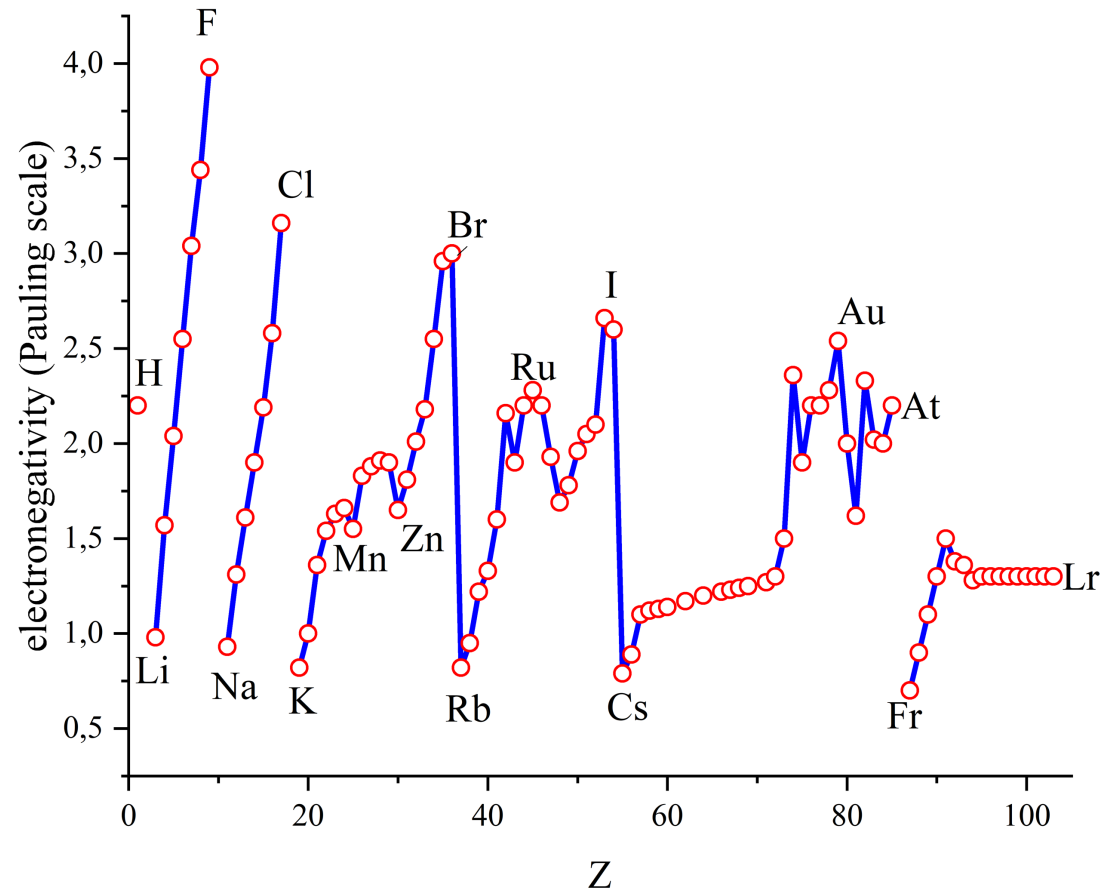
*n*-pentane  
boiling point: 36 °C



# Like dissolves like

**EXERCISE:** Which of these is best dissolved in gasoline (octane,  $C_8H_{18}$ )?

- a.  $H_2O$
- b.  $CH_3CH_2CH_3$
- c.  $CH_3CH_2OH$



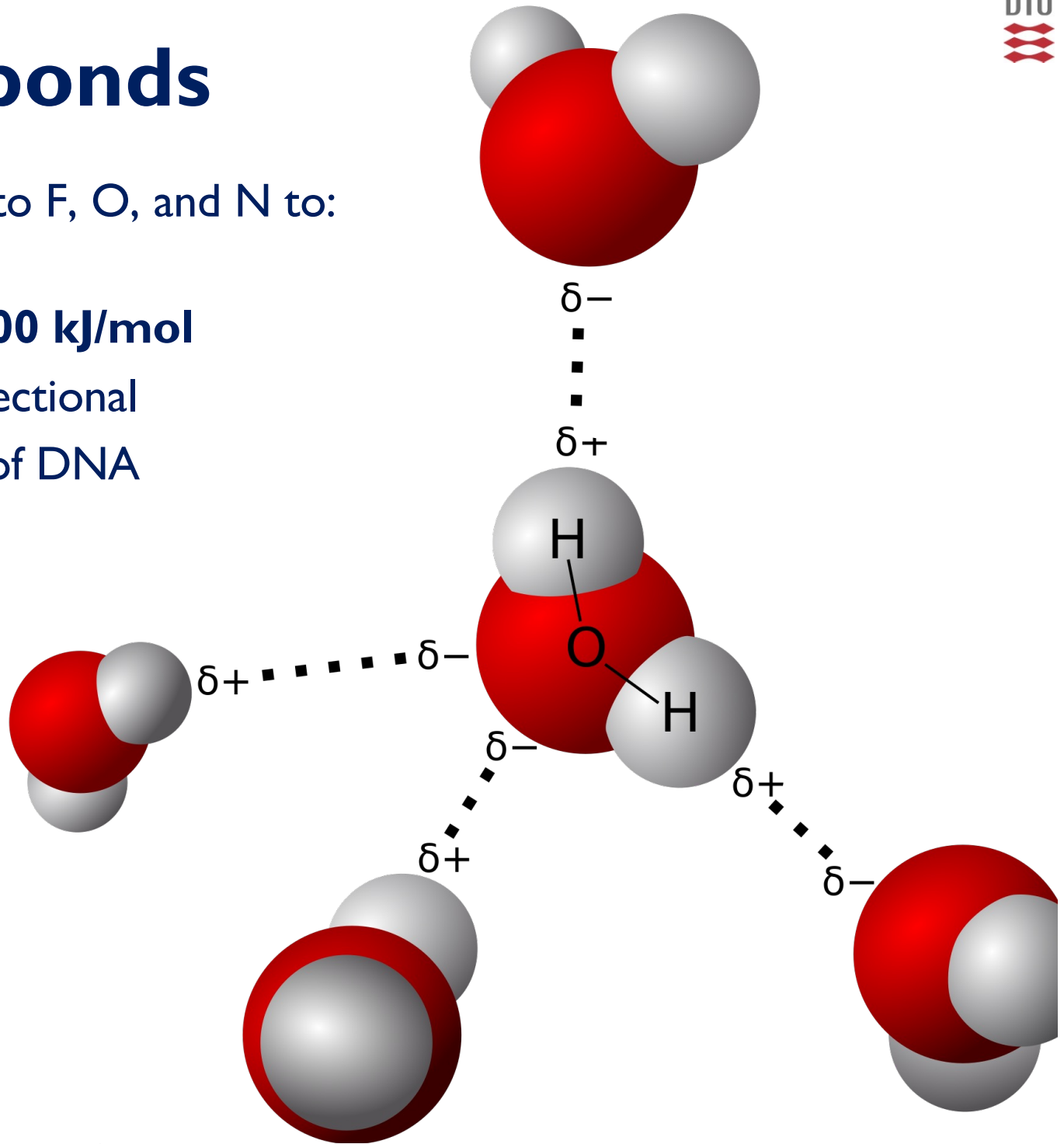
# Hydrogen bonds

Only from H bonded to F, O, and N to:  
F, O, and N.

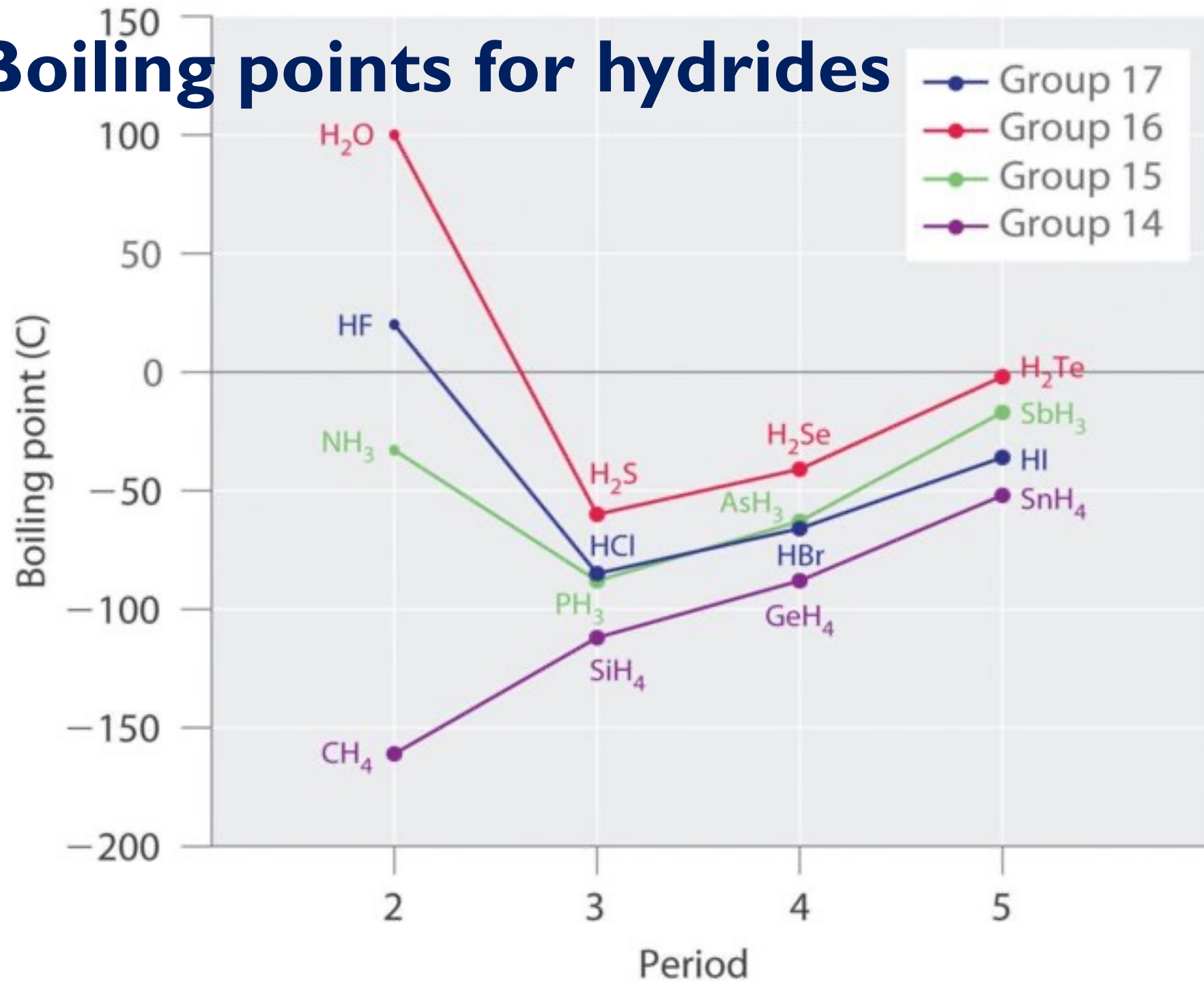
**Bond energy ~10-100 kJ/mol**

Very localized and directional

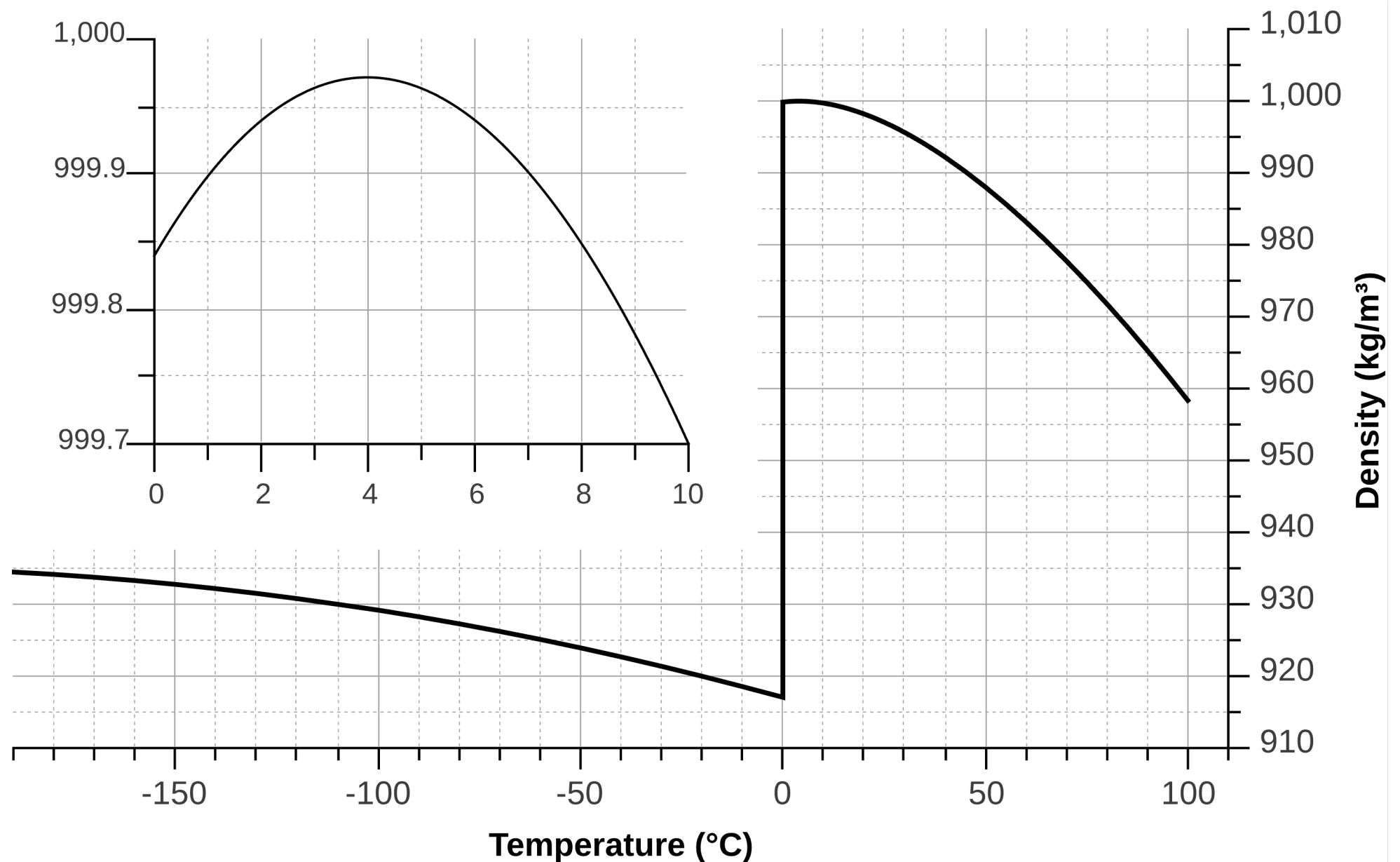
*E.g.:* water, structure of DNA



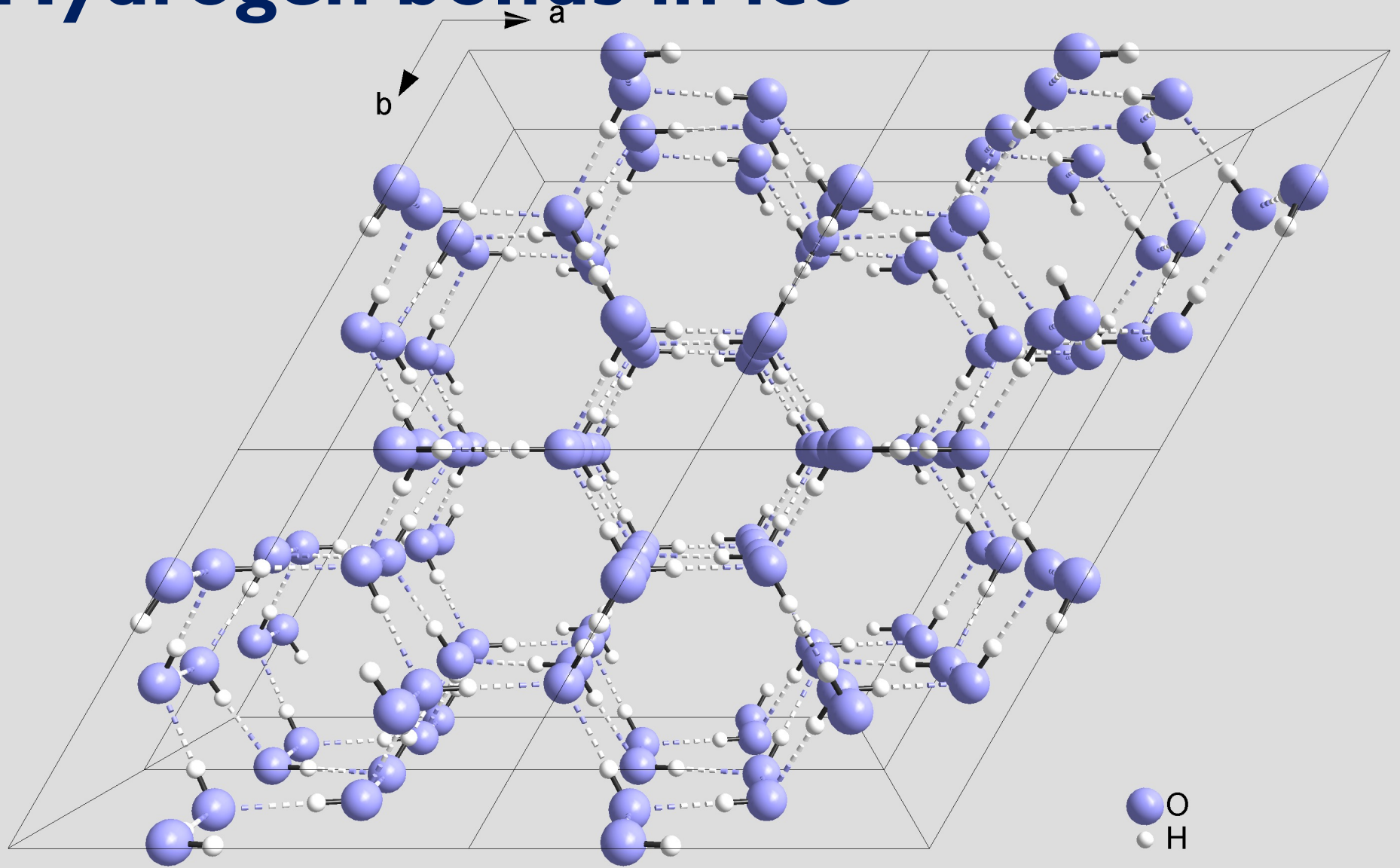
# Boiling points for hydrides



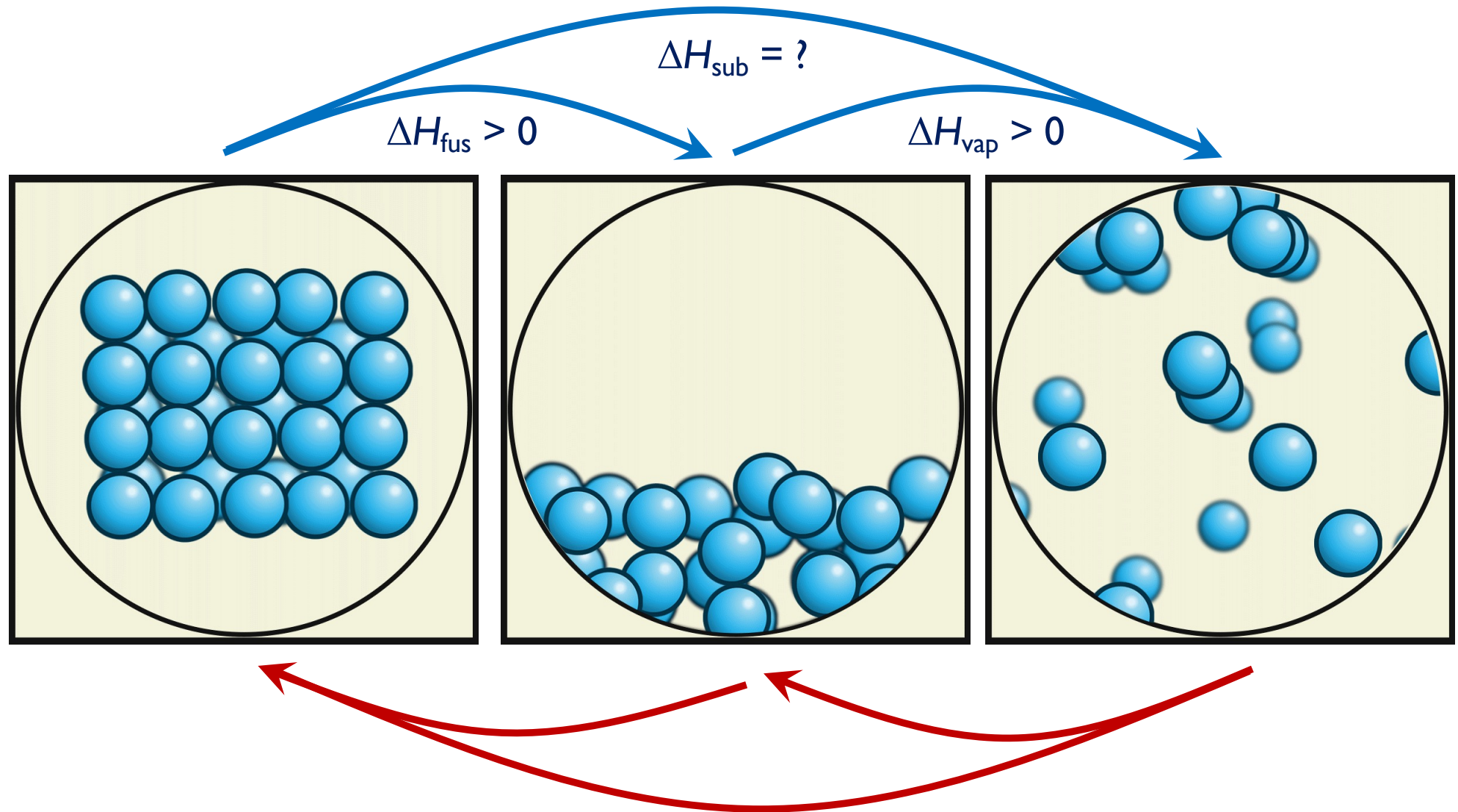
# Hydrogen bonds in ice



# Hydrogen bonds in ice

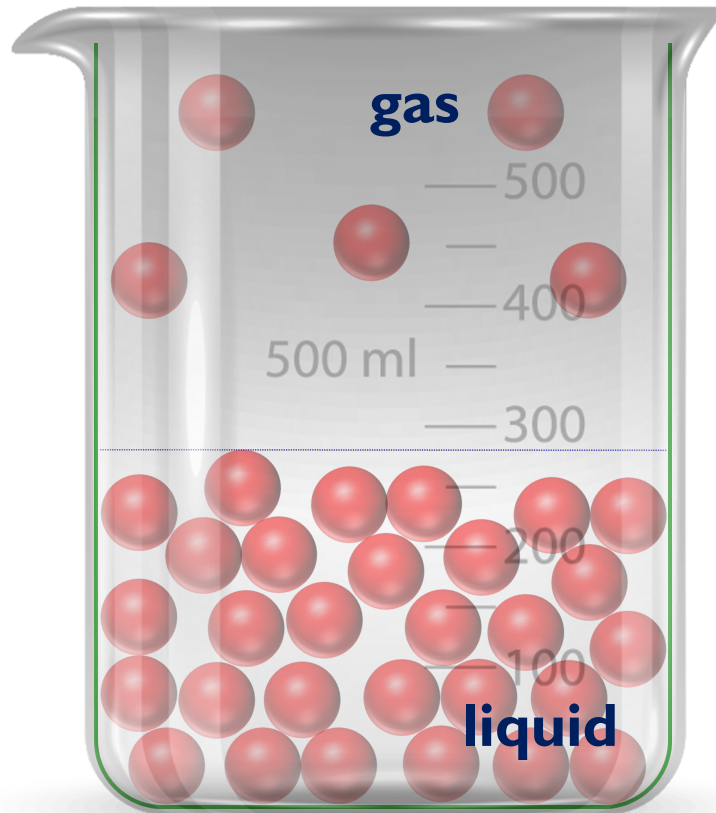


# Phase & phase transitions



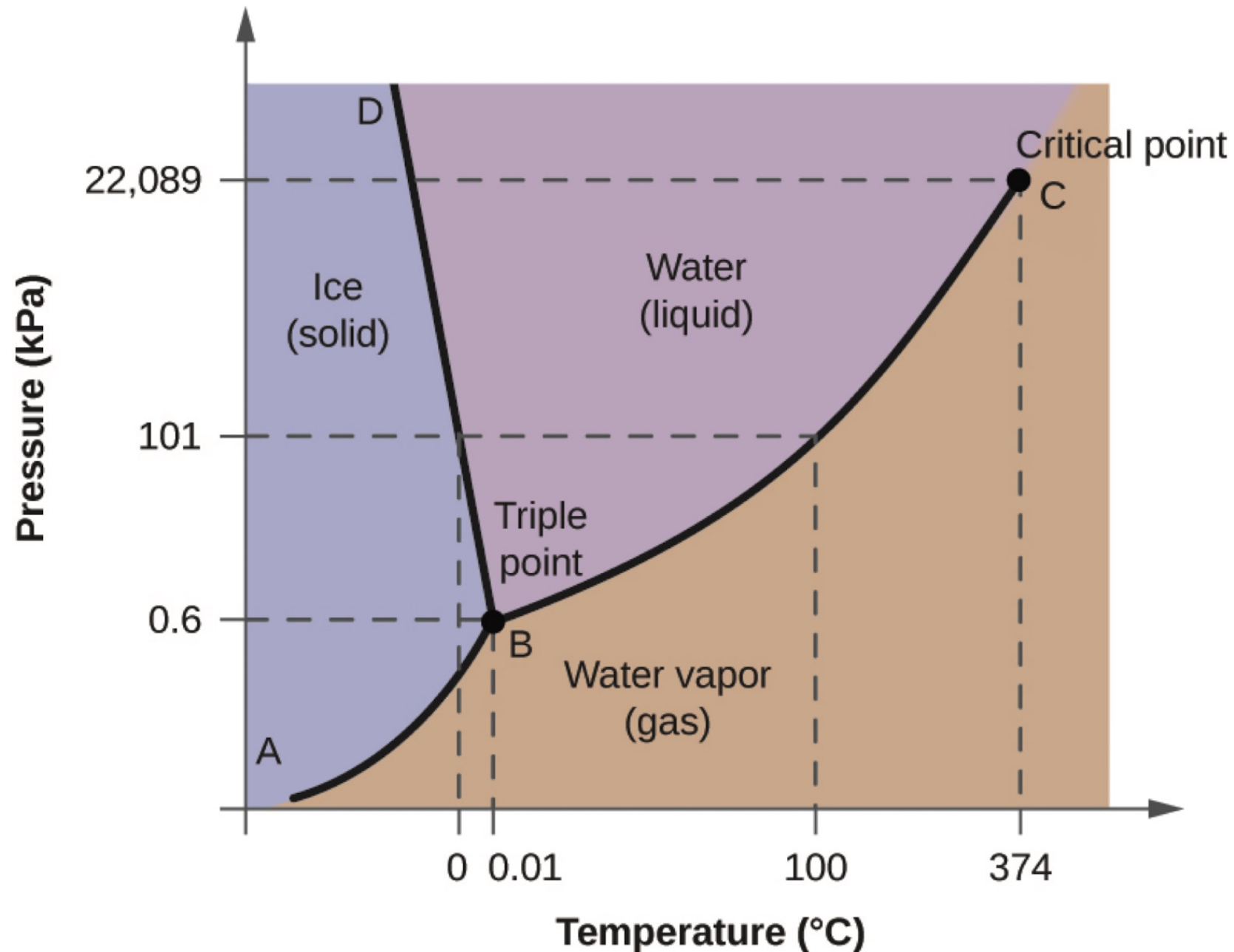


# Vaporization & vapor pressure



| Boiling point of water |          |                                       |                             |
|------------------------|----------|---------------------------------------|-----------------------------|
| location               | height/m | $T_{\text{boiling}}/^{\circ}\text{C}$ | $p_{\text{atm}}/\text{bar}$ |
| Mt. Everest            | 8848     | 70                                    | 0.320                       |
| Mont Blanc             | 4806     | 87                                    | 0.550                       |
| Himmelbjerget          | 147      | 99.5                                  | 0.996                       |
| Death Valley           | -282     | 100.3                                 | 1.027                       |
| Copenhagen             | 2        | 100.0                                 | 1.013                       |

# Phase diagram for water



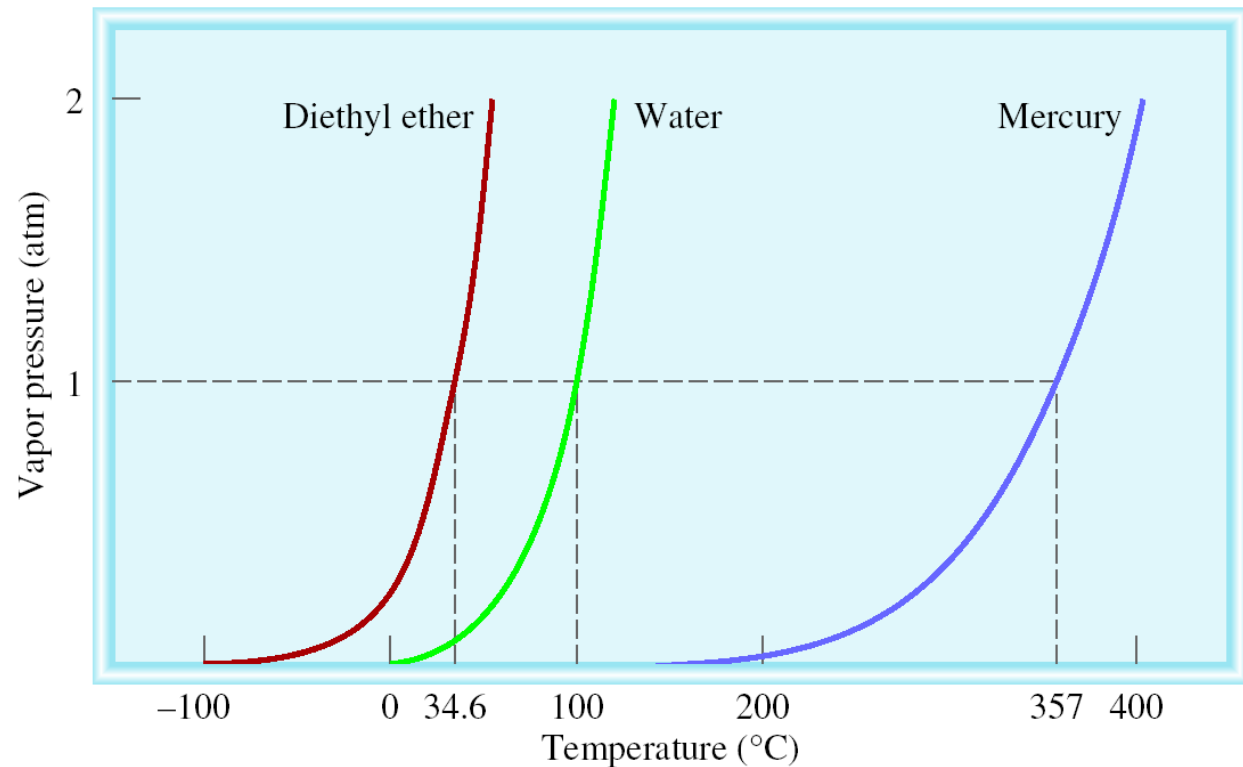


# $\Delta H_{\text{vap}}$ and boiling point

The vapor pressure,  $P$ , of a liquid follows the Clausius-Clapeyron equation

$$\ln(P) = -\Delta H_{\text{vap}}/(RT) + C$$

where  $C$  is a constant and the other symbols have their usual meaning.



**EXERCISE:** Diethyl ether has a vapour pressure of 0.53 bar at 18°C and a  $\Delta H_{\text{vap}}$  of 26.0 kJ mol<sup>-1</sup>.

- What is the vapor pressure at 29°C?
- What is the boiling point at 1 bar?

# Colligative properties:

## Vapor-pressure lowering

If a *solute* is non-volatile (i.e. “no” vapor pressure), the vapor pressure of its *solution* ( $P_1$ ) is lower than that of the pure solvent ( $P_1^\circ$ ).

**Raoult's law:**  $P_1 = X_1 P_1^\circ = (1 - X_2) P_1^\circ$  (for two components)

$X_1$  is the mole fraction of the *solvent* in the solution

$X_2$  is the mole fraction of the *solute* in the solution

### Why?

1) Vaporization increases disorder (entropy). A solution is more disordered than the pure solvent and solute molecules, thus tendency to vaporization is suppressed.

2) Solvent molecules are replaced with solute molecules at the surface.

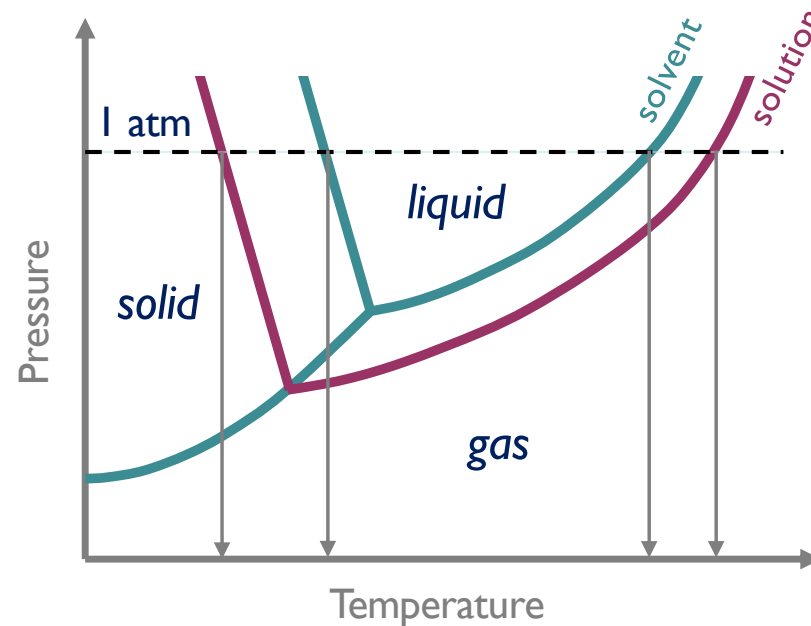
# Colligative properties:

## Boiling point elevation & freezing point depression

### Freezing point depression:

$$\Delta T_f = K_f m$$

where  $K_f$  is a solvent specific constant and  $m$  is the molality.



### Boiling point elevation:

$$\Delta T_b = K_b m$$

where  $K_f$  is a solvent specific constant and  $m$  is the molality.

### Why molality?

Unlike the *molarity* ( $\text{mol L}^{-1}$ ) the *molality* ( $\text{mol kg}^{-1}$ ) is independent of temperature